GLOSSVIS: Visual Glossary Annotations for Scientific Papers

Category: Research

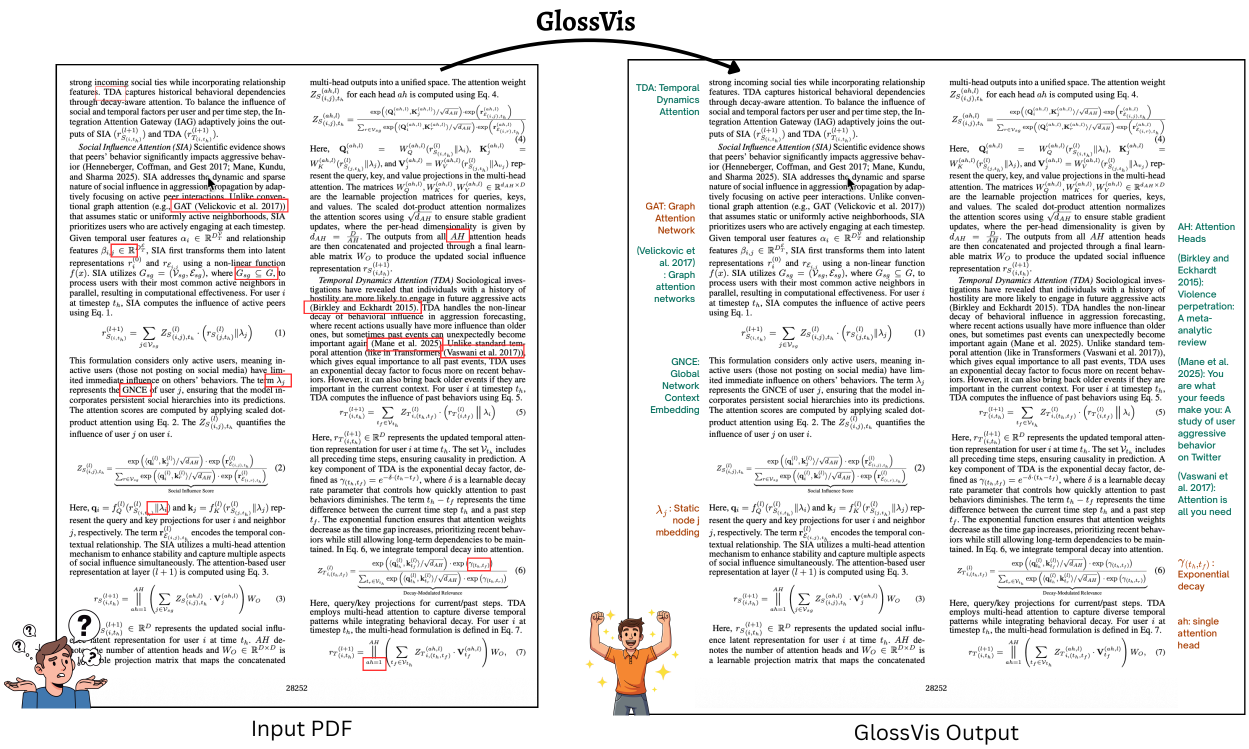

Fig. 1: GLOSSVIS takes an unmodified scientific PDF (left) and produces an annotated version (right) where margin annotations explain citation references, abbreviations, and mathematical symbols in-place. A calibrated two-tier color encoding communicates certainty: ■ for author-defined terms (HIGH confidence) and ■ for model-inferred terms (MEDIUM confidence).

Abstract—Reading dense scientific papers often requires frequent context switching to resolve abbreviations, trace citations, and interpret symbols. This process fragments attention and increases cognitive load for students, practitioners, cross-domain researchers, and even domain experts. Existing AI-based solutions allow users to chat with the scientific papers and provide focused summaries. While these tools ease the cognitive load of a researcher for the horizontal expansion on any subject matter, reading papers is often unavoidable for vertical understanding of the research topic. In the latter objective, the existing tools either fail to integrate implicit glossary information directly into the reading experience or hallucinate paper-specific terminology and do not communicate annotation reliability. We present GLOSSVIS (**Gloss**ary **Vis**ualization), an in-document system that embeds a visual glossary directly within PDF margins, covering citation resolution, abbreviation expansion, and symbol definition at the point of reading. GLOSSVIS combines document-grounded deterministic extraction with retrieval-augmented LLM inference and encodes annotation reliability using a calibrated two-tier confidence scheme. A detailed evaluation across diverse domains shows that GLOSSVIS achieves up to 2× higher exact-match accuracy and 3× lower calibration error than baseline methods, while processing each term offline in 1.1 seconds on average. A user study further shows that 62% of participants prefer GLOSSVIS over a standard PDF, with significantly reduced cognitive load and improved comprehension (p<0.001). Source code is publicly available at GLOSSVIS.

Index Terms—Information visualization, visual analytics, natural language generation

1 INTRODUCTION

Reading a scientific paper is not a purely linear process. Even for experienced researchers, comprehension is repeatedly interrupted by implicit glossary elements such as abbreviations, citation markers, and mathematical symbols whose meanings are defined elsewhere in the document or assumed as prior knowledge [8, 28, 33]. These interruptions introduce frequent context switches, forcing readers to navigate between sections, external tools, or memory, thereby fragmenting attention and increasing cognitive load. Prior work in cognitive load theory and multimedia learning shows that such context switching degrades comprehension and increases mental effort, particularly when information is spatially or temporally separated [27, 37, 39, 43]. As research becomes increasingly interdisciplinary, this challenge intensifies: identical abbreviations often carry domain-specific meanings, and subtle notational (e.g., symbols) differences can alter interpretation, creating ambiguity even for expert readers.

Existing tools address only parts of this problem rather than providing a unified solution. Summarization systems such as ChatPDF[1] and SciSpace[2] generate document-level summaries in a separate interface, requiring readers to leave the primary reading context. LLM-based

[1] https://www.chatpdf.com
[2] https://typeset.io

tools such as ExplainPaper[3] support ad hoc queries but do not provide persistent, in-place annotations or communicate the reliability of their outputs. Reference managers such as Zotero[4] and Mendeley[5] handle citation organization but do not support abbreviations or symbols within the document itself. Consequently, current tools limited to integrate implicit glossary information directly into the reading experience.

Recent advances in LLMs and conversational interfaces have made it easy to query unfamiliar terms during reading [1,7,50]. However, these systems fundamentally rely on global statistical knowledge rather than document-specific grounding. As a result, they often produce fluent but incorrect interpretations (hallucinations) when handling paper-specific terminology [22]. For example, an abbreviation such as `AGI` may be interpreted as "Artificial General Intelligence," despite being explicitly defined as "Aggression Intensity" within a given paper [32]. Retrieval-augmented generation (RAG) partially mitigates this issue by incorporating external context [29]. However, existing systems rarely restrict retrieval to the document being read and fail to present uncertainty in an interpretable manner. These limitations highlight the need for approaches that integrate contextual information directly within the reading artifact while preserving usability. Research on augmented and fluid documents explores embedding interactive elements within text [10], while systems for active reading support highlighting and note-taking [8]. However, these approaches rely on document reformatting, external interfaces, or manual interaction. Visualization research emphasizes the importance of spatial proximity and integrated representations for reducing cognitive effort [26, 34]. At the same time, uncertainty visualization studies highlight the need to communicate the reliability of derived information, especially when automated systems are involved [9, 20]. Despite these advances, integrating implicit glossary information directly within documents while explicitly communicating its reliability remains underexplored.

To address this gap, we introduce GLOSSVIS, an in-document visualization system that augments scientific PDFs with margin-based glossary annotations. Given a PDF paper, GLOSSVIS detects and annotates three implicit glossary elements, such as citations, abbreviations, and mathematical symbols, and renders their resolution directly alongside the text at the point of reading. The system adopts a hybrid extraction strategy that prioritizes deterministic information extraction from the document itself and falls back to LLM with retrieval-augmented generation when definitions are absent. This design ensures that author-defined meanings are preserved while maintaining coverage for implicitly used terms. GLOSSVIS also visually encodes whether an annotation is derived from explicit document evidence or inferred through LLM-based reasoning. We evaluate GLOSSVIS on a ground-truth benchmark of papers spanning diverse domains, along with a human study involving nine evaluators across multiple documents. GLOSSVIS achieves up to 2× higher exact-match accuracy and 3× better calibration (ECE = 0.258 vs. 0.763) than comparable LLM baselines, with an average processing cost of 1.1 s per term. The human study shows that 62% of participants prefer GLOSSVIS over a standard PDF, with significantly reduced cognitive load and improved comprehension ($p<0.001$).

In summary, our key contributions are as follows:

- We introduce an in-document visualization approach for annotating implicit glossary information (citations, abbreviations, and symbols) directly within scientific PDFs.
- We propose a hybrid extraction pipeline that combines deterministic document-grounded methods with LLM retrieval-based inference to balance accuracy and coverage.
- We design a confidence-aware visual encoding that makes the reliability of automatically generated annotations explicit to the reader.
- We demonstrate the effectiveness of the approach through quantitative and qualitative evaluation across diverse scientific domains.
- We release GLOSSVIS as an open-source tool to support reproducibility and future research.

[3]https://www.explainpaper.com

[4]https://www.zotero.org

[5]https://www.mendeley.com

2 Related Work

A growing body of work has explored how computational methods and interactive systems can support the reading and understanding of scientific documents. Early work focused on manual annotation and note-taking to aid comprehension and knowledge organization [33]. More recent systems have explored augmented reading environments, including structured document transformations and fluid documents that embed interactive elements directly within text [10]. Active reading systems further support highlighting, summarization, and explanation during reading [8]. Systems such as ScholarPhi [18] provide position-aware definitions of terms and symbols through an interactive PDF overlay. However, these approaches do not provide automated, in-place augmentation of unified implicit glossary information within the document without reliability.

Recent advances in large language models (LLMs) have enabled AI-assisted reading and explanation systems. Conversational interfaces and document-based question answering systems allow users to query unfamiliar terms on demand [1, 7, 50]. Retrieval-augmented generation (RAG) improves factual grounding by incorporating external context [29]. At the same time, work at the intersection of visualization and LLMs highlights both opportunities and limitations. It shows that LLMs struggle with structured interpretation and require human-interpretable representations for effective interaction [4, 6, 24]. Prior work in uncertainty visualization shows that visual encodings affect user interpretation and trust [9, 20]. Similarly, LLM-assisted systems highlight the need to expose model confidence to prevent over-reliance and support informed decisions [11, 30]. Despite these advances, most systems operate outside the document context and require users to switch interfaces. They may also produce hallucinated or contextually incorrect interpretations when document-specific meaning is required [22].

Pattern-based information extraction approaches identify abbreviation-definition pairs from text [42]. More recent models incorporate domain-specific language processing and layout-aware representations for improved document understanding [35, 47]. Paper Plain [3] system extract surface definitions and simplify technical jargon. Visualization-oriented systems have explored integrating extraction pipelines with analytical workflows [49]. Although these methods are not designed for direct integration into reading interfaces and do not address how extracted knowledge should be presented to support comprehension. Visualization research has emphasized the importance of spatial proximity and integrated representations for reducing cognitive effort [26, 34]. Recent work extends these ideas to document-centered visual analytics, enabling interactive exploration of document structure and content [14, 36]. However, most approaches operate at the corpus level or rely on external visualizations, rather than augmenting the document itself as the primary interaction medium.

Our work integrates these aspects into a unified, in-document visualization that embeds implicit glossary information while clearly conveying its reliability.

3 Task Abstraction

GLOSSVIS is motivated as a visualization problem rather than purely an engineering task. Following Munzner's nested model [34], we frame scientific reading as the task of resolving *implicit glossary elements* without leaving the primary reading context.

The domain-level task is *to enable a reader to resolve implicit knowledge references in a scientific paper without leaving the primary reading artifact*. Three types of implicit references recur across virtually every paper in computer science and related fields. **Citation references** (`[14]`, `Smith et al., 2020`) are numeric or author–year markers that refer the reader to bibliography entries containing details such as title, authors, and year. **Abbreviations** (CNN, GAT, TSGAN, AGI) are all-caps tokens whose expansions may be defined inline, omitted entirely, or assigned paper-specific meanings that differ from common usage. **Symbols** (λ, ∇_θ, $\mathscr{L}$) are mathematical operators or variables typically defined once and reused throughout the document. The unifying structure of all three types is the same. A compact token in the text body carries meaning that is defined elsewhere, such as in a reference list, an earlier paragraph, or the reader's possibly incomplete or incorrect prior knowledge. Resolving such references requires a context switch that interrupts the reading flow.

At the abstract level, annotating a scientific document for seamless reading requires three operations for each implicit reference. (i) Locate: detect the compact token in the text stream and determine its type (citation, abbreviation, or symbol). (ii) Identify: retrieve or infer its expansion or meaning. (iii) Contextualize: communicate both the meaning *and* its reliability to the reader at the point of encounter. The *Contextualize* operation is often omitted by existing tools. Similarly, an LLM chat interface supports *Identify* on demand but provides neither a reliability signal nor spatial proximity.

Further, we derive five design requirements that any annotation system for this task must satisfy. *(R1) Spatial proximity* places annotations at the point of reading, within the foveal or parafoveal field of the trigger token, so that no saccadic context switch is required. *(R2) Confidence visibility* makes the reliability of each annotation explicit within the reading artifact, so readers can distinguish author-stated definitions from model inferences without leaving the document. *(R3) Domain fidelity* ensures that annotations for author-defined terms reflect the paper's own definitions rather than generic model knowledge that may conflict with paper-specific meaning. *(R4) Offline operation* requires the system to function without a network connection or external API keys, making it suitable for air-gapped and bandwidth-limited environments. Furthermore, require on-device processing that maintains data privacy. *(R5) Non-occlusion* prevents annotations from overlapping or obscuring the body text, preserving readability at all zoom levels. These requirements guide the design decisions in §4 and the implementation choices in §5.

4 Design Space and Rationale

We describe the key design decisions underlying GlossVis and justify them through comparison with plausible alternatives. These decisions are guided by the requirements derived from §3, particularly the need for spatial proximity, reliability communication, and domain fidelity.

A central design choice is to place annotations directly in the document margins, spatially adjacent to their trigger tokens, rather than presenting them in a separate side panel. While side panels are common in existing systems, they require users to repeatedly shift their gaze between the reading position and an external view. This introduces additional cognitive load, as readers must maintain spatial context across two locations. In contrast, margin-based annotations satisfy spatial proximity by placing explanations within the foveal or parafoveal field, reducing visual search and reorientation effort. This design is consistent with cognitive load theory [39] and the Gestalt principle of proximity [26], both of which suggest that co-located information improves comprehension. A second design decision concerns how annotation reliability is represented. Instead of continuous confidence scores or binary labels, GlossVis uses three categorical levels: HIGH, MEDIUM, and LOW (discarded). Continuous confidence values are difficult for most readers to interpret, particularly without calibration knowledge, while binary labels are not feasible at inference time due to the absence of ground truth. A three-level scheme provides a clear mapping to user action: HIGH-confidence annotations can be trusted, MEDIUM-confidence annotations may require verification, and LOW-confidence outputs are not shown. This design balances interpretability and expressiveness, and aligns with prior findings in uncertainty communication [21]. The use of color encoding (green and orange) further reinforces this distinction while remaining accessible to colorblind users. A third key decision is to prioritize Deterministic Information Extraction (DIE) over model-based inference. GlossVis adopts an DIE-first strategy, where regex-based extraction is used to identify author-defined terms directly from the document, and LLM-based retrieval is used only as a fallback. This contrasts with LLM-first approaches that rely primarily on model inference. Empirical results show that zero-shot LLMs perform poorly on paper-specific abbreviations, as these terms often do not appear in training data. In contrast, DIE reads the authors' own definitions, providing structural correctness when definitions are present. This ensures domain fidelity and avoids introducing hallucinated interpretations in the highest-confidence annotations. Finally, we employ a structural validation mechanism to filter LLM-generated outputs. Instead of relying on self-reported confidence scores, which are often miscalibrated [16], we apply an Initial-Letter Validation (ILV) check that verifies whether the initials of an expansion match the corresponding abbreviation. This deterministic filter removes a common class of hallucinations, plausible but structurally inconsistent expansions, without requiring probabilistic calibration. As a result, the system maintains reliable confidence separation, with low-confidence outputs discarded prior to rendering.

Formal Definitions

We formalize the core components of GlossVis to characterize the extraction, validation, and rendering pipeline. Let a denote an annotation with resolution source $s(a) \in \{\text{DIE}, \text{RAG}, \text{DISCARD}\}$. The assigned confidence score $P(a)$ is defined in equation 1.

$$P(a) = \begin{cases} 0.95 & \text{if } s(a) = \text{DIE} \\ 0.65 & \text{if } s(a) = \text{RAG (passes ILV)} \\ 0 & \text{if } s(a) = \text{DISCARD} \end{cases} \tag{1}$$

The value $P(a) = 0.95$ reflects the near-perfect empirical precision of DIE-based extraction on inline-defined terms, while remaining below 1.0 to avoid overconfident calibration. The value $P(a) = 0.65$ corresponds to the empirically observed accuracy of RAG-based expansions after structural validation (§8.4). Annotations that fail validation or yield empty outputs are assigned zero confidence and are not rendered.

Extraction follows a hybrid strategy. For a glossary element t in document D, the predicted expansion $\hat{e}(t)$ is defined in equation 2.

$$\hat{e}(t) = \begin{cases} \text{DIE}(t, D) & \text{if DIE}(t, D) \neq \emptyset \\ \text{LLM}(t,\ \text{Retrieve}(t, D, k)) & \text{otherwise} \end{cases} \tag{2}$$

Here, Retrieve(t,D,k) returns the top-k relevant chunks (k=2) from a FAISS index over D. The index uses 500-character segments with a 100-character overlap and `all-MiniLM-L6-v2` embeddings [48]. To validate candidate expansions, we use a fuzzy initial-matching criterion. Let $w = w_1 w_2 \cdots w_m$ be a candidate phrase, after removing function words $\mathscr{F} = \{\text{a, an, the, of, for, in, on, at, to, by, and, or, with, from, via}\}$. Let $I(w)$ denote the sequence of initials of the remaining words. A phrase w is accepted as a valid expansion of an abbreviation A if $I(w)$ forms a subsequence of A and $|,|I(w)| - |A|,| \leq 1$. This constraint eliminates structurally inconsistent expansions while tolerating minor tokenization differences. We evaluate confidence calibration using Expected Calibration Error (ECE), as defined in Equation 3, which measures the mismatch between predicted confidence and actual accuracy.

$$\text{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left| \text{acc}(B_b) - \text{conf}(B_b) \right|, \tag{3}$$

Here, $B = 5$ denotes the number of equal-width confidence bins, acc(B_b) represents the empirical accuracy, and conf(B_b) denotes the mean predicted confidence for bin b. We also report Token-F1 to capture partial correctness, as defined in Equation 4.

$$\text{TF1}(\hat{e}, e^) = \frac{2 \cdot |\hat{T} \cap T^*|}{|\hat{T}| + |T^*|}, \tag{4}$$

Here, $\hat{T}$ and T^* denote the token sets of the predicted and ground-truth expansions after normalization. Token-F1 provides partial credit for near-correct expansions (e.g., "Globally Average-pooling" vs. "Global Average Pooling").

Finally, spatial placement follows a non-overlap constraint. An annotation with bounding box B_i is rendered only if $B_i \cap B_j = \emptyset$ for all previously placed annotations on the same page and margin. If an overlap occurs, the placement is skipped and deferred to the next valid occurrence. This ensures that annotations do not occlude each other and preserves document readability.

5 GLOSSVIS: GLOSSARY VISUALIZATION SYSTEM

Fig. 2 illustrates the GLOSSVIS pipeline. Given an unmodified input PDF, the system produces a margin-annotated PDF and a sidecar JSON file, all without requiring a network connection (R4). The pipeline proceeds in three phases. **Phase 1** (Glossary Element Detection) processes the document through two parallel tracks to extract candidate glossary elements. Plain-text (Unicode) extraction identifies abbreviations, citations, and inline symbols, while equation-region OCR identifies symbols not recoverable from Unicode text. **Phase 2** (Glossary Resolution) resolves each candidate into a human-readable gloss. It first applies Deterministic Information Extraction (DIE) and falls back to a local LLM-RAG pipeline only when the document does not contain an explicit definition. **Phase 3** (Spatial Margin Rendering) places each resolved annotation in the nearest page margin, enforcing non-overlap through a dry-run collision check, and serializes the results to a sidecar JSON file for the interactive viewer. The resolution path determines both the confidence tier and visual encoding. DIE-resolved annotations are **green**, while RAG-resolved annotations are **orange**. Low-confidence RAG outputs are discarded.

5.1 Glossary Element Detection

GLOSSVIS identifies candidate glossary elements by parsing each page of the input PDF using `PyMuPDF` [6], which provides a structured representation of text blocks with precise spatial coordinates. This enables joint access to textual content and layout information required for in-document annotation. Detection is performed on the original document and proceeds through two complementary tracks: one targeting abbreviations and citations from plain-text spans, and the other targeting mathematical symbols from both Unicode text and equation regions. For equation blocks that cannot be reliably recovered through text extraction, the system additionally employs `LatexOCR` [7] to obtain LaTeX representations.

5.1.1 Abbreviation and Citation Detection

This track works on the plain-text extraction of the document to identify abbreviation and citation candidates. Abbreviations are detected by scanning each text span for Isolated uppercase tokens of length 2–8 are detected using the regex P_{abbr}, defined as `(?<!\w)[A-Z][A-Z0-9\-]1,5(?!\w)`. To reduce noise from incidental tokens (e.g., figure labels or headings), a candidate is retained only if it appears at least twice in the document. Once annotated, a term is re-annotated only if it reoccurs more than five pages after its previous occurrence, balancing readability and redundancy. Further, citation detection supports both numeric and author–year formats commonly used in scientific writing. Numeric citations (e.g., [18], [3,,7]) are identified using two regex patterns: $P_{\text{num}} =$ `\[\s*(\d1,3)\s*\]` and $P_{\text{cluster}} =$ `\[\s*(\d1,3(?:\s*,\s*\d1,3)+)\s*\]`. Author–year citations (e.g., Harrison et al., 2014) are detected using the regex P_{ay}, defined as `\b([A-Z][A-Za-z\'\-]+)\s+et\s+al\.\s*((?:19|20)\d2[a-z]?)`, with analogous patterns for multi-author cases. To resolve citations, the system performs a backward scan to locate the References section and constructs an index of entries. Numeric citations are mapped by integer keys, while author–year citations are normalized into canonical strings, enabling constant-time lookup during resolution. Each citation is annotated only at its first occurrence in the main text.

5.1.2 Mathematical Symbol Detection

This track targets mathematical symbols, which often appear either as Unicode characters in text or as rasterized equation images. To handle both cases, GLOSSVIS employs two complementary detection passes. The first pass scans text spans for contiguous Unicode characters within mathematically relevant ranges using the regex P_{sym}, defined as `[\u0370-\u03FF \u2200-\u22FF \u2A00-\u2AFF \u2070-\u209F]+`. These ranges cover Greek letters, standard mathematical operators, extended operators, and super/subscript characters. Common structural symbols such as $=$, $+$, and $\leq$ are excluded using a fixed filter, as they do not convey domain-specific meaning. The second pass detects symbols embedded in equation images that are not recoverable through text extraction. A text block is classified as an equation region if it contains fewer than 12 alphabetic words and exhibits at least two of the following features: relational operators (e.g., $=$, $<$, $>$), summation or product operators (e.g., $\sum$, $\prod$, $\sqrt{\,}$), sub/superscripts, or bracket delimiters. Identified regions are rasterized at 200 DPI and processed using `LatexOCR` to obtain a LaTeX representation. After filtering formatting commands (e.g., `\text`, `\frac`), the remaining tokens are added to the symbol candidate set.

5.2 Glossary Resolution

Glossary resolution maps each detected element to a human-readable gloss by prioritizing document-grounded evidence over model-based inference. GLOSSVIS follows a deterministic-first strategy, where definitions are first extracted directly from the document and only unresolved cases are forwarded to a retrieval-augmented language model. This design ensures that author-defined meanings are preserved whenever available, while maintaining coverage for implicit or undefined terms.

5.2.1 Abbreviation Resolution

Abbreviations are resolved using a deterministic-first strategy that prioritizes in-document definitions. Following the approach by [42], the system scans the document for inline patterns of the form "Full Form (ABBR)" using the regex $P_{\text{def}} =$ `((?:[A-Za-z\-]+\s+)1,10)\(([A-Z]2,8)\)`. To ensure robustness on typeset PDFs, preprocessing normalizes Unicode ligatures (e.g., `fi`, `fl`) to their ASCII equivalents and rejoins hyphenated line breaks, enabling accurate recovery of definitions split across lines. Each candidate expansion is validated using an initial-letter matching criterion. Let $W = [w_1, \ldots, w_k]$ denote the sequence of words in a candidate phrase after removing function words $\mathscr{F} = \{$a, an, the, of, for, in, on, at, to, by, and, or, with, from, via$\}$. The initial sequence $\mathscr{I}(W)$ is formed by taking the first letter of each remaining word. A phrase is accepted if $\mathscr{I}(W)$ matches the abbreviation exactly, with a relaxed condition allowing a one-character deviation to accommodate compound-word variations. Abbreviations resolved through this deterministic process are assigned high confidence and rendered accordingly. If no inline definition is found, the system falls back to a retrieval-augmented language model. Instead of querying with the abbreviation alone, GLOSSVIS uses the surrounding sentence context to retrieve relevant document chunks, improving contextual grounding. The generated expansion is then subjected to the same initial-letter validation; only structurally consistent outputs are retained. Valid results are assigned medium confidence, while inconsistent or empty responses are discarded.

5.2.2 Citation Resolution

Citation resolution leverages the reference index constructed during detection (5.1.1). Numeric citations are resolved through direct integer lookup, while author–year citations are matched using normalized author–year keys. Both cases rely on deterministic matching against the reference list, ensuring that bibliographic information is recovered exactly as defined in the document. For each resolved citation, the system extracts key metadata such as title and year from the indexed entry and generates a concise gloss. These annotations are assigned high confidence, as they are derived directly from the document structure without inference. In cases where author–year entries cannot be reliably matched due to formatting inconsistencies or parsing ambiguities, the system falls back to a language model. The model is provided with the raw reference string and returns structured metadata, which is validated against the original entry before annotation. Such cases

[6] https://pymupdf.readthedocs.io/en/latest/
[7] https://lukas-blecher.github.io/LaTeX-OCR/

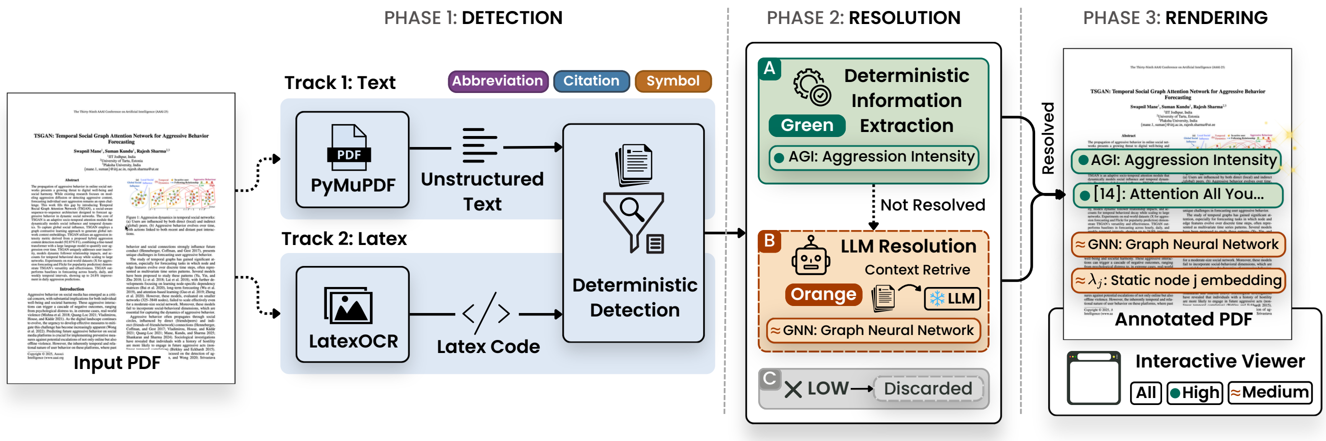

Fig. 2: **GlossVis system pipeline.** The system processes an input PDF in three phases. Phase 1 extracts candidate glossary elements. Phase 2 resolves them using deterministic matching, with fallback to a local LLM-RAG pipeline for unresolved cases. Phase 3 renders annotations in the page margins and encodes confidence through color.

are assigned medium confidence, while outputs that are unresolved or contain inconsistent metadata (e.g., an extracted year that is not a valid four-digit integer) are discarded.

5.2.3 Symbol Resolution

Resolving mathematical symbols is more challenging than abbreviations or citations, as their meanings are rarely expressed in a consistent or explicit format. Instead, symbols are typically defined through surrounding prose, equation context, or figure captions. To address this, GlossVis adopts a deterministic-first strategy that prioritizes in-document evidence before resorting to model-based inference. The system first attempts rule-based resolution by identifying common definitional patterns inspired by prior work on mathematical definition extraction [2]. These include constructions such as P_{where} and P_{let}, defined as `where;`*sym*`;(is|denotes?|represents?);(.+)` and `[Ll]et;`*sym*`;(be|denote|equal);(.+)`, respectively. along with caption-local definitions and preceding-sentence contexts. Extracted candidate meanings are filtered to ensure they are non-empty, concise, and do not trivially repeat the symbol itself. Symbols resolved through this deterministic process are assigned high confidence, as they are grounded directly in the document. When no valid in-document definition is found, the system falls back to the retrieval-augmented formulation defined in Eq. 2. The language model is provided with the symbol (in Unicode or LaTeX form) along with a local context window surrounding its first occurrence. It returns a short textual description of the symbol's meaning. Since these outputs rely on inference rather than explicit definitions, they are assigned medium confidence. Low-confidence or empty responses are discarded during post-processing.

5.3 Spatial Margin Rendering

The rendering stage integrates resolved glossary annotations directly into the document by allocating dedicated margin space and placing annotations in spatial alignment with their trigger tokens. GlossVis operates on a scaled copy of the original PDF, creating blank margins while preserving the original content layout. The system supports both single-column and two-column documents without explicit layout detection by assigning each text block $b = (x_0^b, y_0^b, x_1^b, y_1^b)$ to a margin side based on its horizontal midpoint (Eq. 5).

$$c(b) = \begin{cases} 1 & \text{if } \dfrac{x_0^b + x_1^b}{2} < \dfrac{W_0}{2} \\ 2 & \text{otherwise} \end{cases} \tag{5}$$

This rule places annotations adjacent to their source column in multi-column layouts and distributes them across both margins in single-column documents. Annotations are then positioned using a collision-aware layout that respects document structure and avoids visual overlap.

5.3.1 Page Scaling and Margin Creation

GlossVis expands each page horizontally to accommodate in-document annotations without altering the original content. Let W_0 and H_0 denote the original page width and height. Each page is scaled by a factor $s = 1.2$ and centered on a new canvas, as shown in Equation 6.

$$W' = s \cdot W_0, \qquad x_{\text{off}} = \frac{(s-1)\,W_0}{2} \tag{6}$$

Here, W' is the width of the scaled page and x_{off} is the horizontal offset used to center the original content. This transformation introduces blank margins on both sides while preserving the layout of the source document. Typeset PDFs often contain internal whitespace beyond their nominal bounds. The system therefore computes a content-aware bounding box for each page and projects it into the scaled coordinate system, as defined in Equation 7.

$$B_p^{\text{content}} = \left(x_p^{(0)} + x_{\text{off}},\ y_p^{(0)},\ x_p^{(1)} + x_{\text{off}},\ y_p^{(1)}\right) \tag{7}$$

All annotation placement is constrained relative to B_p^{content}, ensuring that annotations respect the true content boundaries rather than idealized page geometry.

Each annotation e is anchored to its source token bounding box $b_e = (x_0^e, y_0^e, x_1^e, y_1^e)$. The vertical anchor position is defined as in Equation 8.

$$y_{\text{anc}}(e) = y_0^e + \delta_y, \qquad \delta_y = 3\,\text{pt} \tag{8}$$

This aligns the annotation with the baseline of the corresponding text line. Let $x_{c,p}^{(0)}$ and $x_{c,p}^{(1)}$ denote the left and right content boundaries from B_p^{content}. The target margin region for annotation placement is then defined as in Equation 9.

$$R_m(e) = \begin{cases} \left[\delta_p,\ y_{\text{anc}}(e),\ x_{c,p}^{(0)} - \delta_p,\ H_0 - \delta_p\right] & c(e) = 1 \\ \left[x_{c,p}^{(1)} + \delta_p,\ y_{\text{anc}}(e),\ W' - \delta_p,\ H_0 - \delta_p\right] & c(e) = 2 \end{cases} \tag{9}$$

Here, $\delta_p = 5\,\text{pt}$ provides padding from the content boundaries. This formulation ensures that annotations are placed within available margin space and can wrap naturally across multiple lines without overlapping the original document content.

Algorithm 1 Annotation Placement with Margin Collision Avoidance

Require: Ordered list $\mathscr{A} = \{(t_i, d_i, p_i, c_i, \text{conf}_i, b_i)\}_{i=1}^{n}$ (term, definition, page, column, confidence, source bbox); scaled document $\mathscr{D}$; per-page content bboxes $\{B_p^{\text{content}}\}$; scaled width W', height H_0, padding δ_p, offset δ_y

Ensure: Placed-annotation set $\mathscr{S}$; updated document $\mathscr{D}$

$\mathscr{S} \leftarrow \emptyset$
for each annotation $a_i \in \mathscr{A}$ **do**
  Compute y_{anc} and $R_m(a_i)$ via Eqs. (8)–(9)
  $\hat{t} \leftarrow \texttt{FormatAnnotation}(t_i, d_i, \text{conf}_i)$ ▷ type icon + "**term**: expansion"
  $\mathscr{D}_{\text{tmp}} \leftarrow$ new single-page document (W', H_0)
  $\texttt{TextWriter.fill\_textbox}(R_m(a_i), \hat{t}, 5\,\text{pt})$; write to $\mathscr{D}_{\text{tmp}}[0]$ ▷ dry-run render
  $B_{\text{ann}} \leftarrow \texttt{GetTextBlocks}(\mathscr{D}_{\text{tmp}}[0])[0]$.bbox; close $\mathscr{D}_{\text{tmp}}$
  if a_i is a symbol with glyph image I **then** $B_{\text{ann}} \leftarrow B_{\text{ann}} \cup B_I$
  end if
  $M_c \leftarrow$ full margin strip for column c_i on page p_i
  $\mathscr{E} \leftarrow \{B \in \texttt{GetTextBlocks}(\mathscr{D}[p_i]) \mid B \subseteq M_c\}$
  if $\exists B \in \mathscr{E}$ s.t. $B_{\text{ann}} \cap B \neq \emptyset$ **then**
    continue ▷ skip occurrence; defer to next unobstructed page
  else
    Write $\hat{t}$ to $\mathscr{D}[p_i]$ with color(conf$_i$), opacity $\alpha(\text{conf}_i)$
    if a_i is a symbol **then** insert glyph I at B_I in $\mathscr{D}[p_i]$
    end if
    $\mathscr{S} \leftarrow \mathscr{S} \cup \{B_{\text{ann}}\}$
  end if
end for
return $\mathscr{S}$, $\mathscr{D}$

5.3.2 Collision-Aware Placement

GlossVis uses a collision-aware placement strategy to ensure readability and avoid visual clutter (R5) by preventing overlap between annotations within the margin. For each candidate annotation, a target margin region is computed using Eq. 9 and anchored to the corresponding text position. Before placing an annotation on the page, the system estimates its rendered extent using a *dry-run render*. Instead of approximating annotation height analytically, which would require replicating font metrics and line-wrapping behavior, the system renders the annotation on a temporary in-memory canvas and retrieves its bounding box. This produces an exact estimate of the space required, including multi-line cases. Let B_{ann} denote the resulting annotation bounding box, and let $\mathscr{E}$ be the set of existing annotation boxes within the same page and margin. The annotation is placed only if $B_{\text{ann}} \cap B = \emptyset$ for all $B \in \mathscr{E}$. If a collision is detected, the annotation is skipped at the current occurrence and deferred to its next valid occurrence in the document.

Algorithm 1 summarizes the complete placement procedure. Each annotation is processed in reading order, and a dry-run render is used to compute its true bounding box before collision checking. When no overlap is detected, the annotation is rendered into the target margin region with visual encoding determined by its confidence level. For symbol annotations, associated glyph images are inserted alongside the text. The set of occupied bounding boxes is updated incrementally, ensuring that subsequent annotations respect existing placements. This design guarantees non-overlapping annotations while preserving spatial alignment with the source text.

5.3.3 Annotation Rendering

Resolved annotations are rendered directly within the allocated margin space using a consistent visual encoding. Each annotation is formatted as a compact text block consisting of a type indicator, the term in bold, and its corresponding expansion or reference. The rendering is performed using `insert_textbox` at a fixed font size. Visual encoding reflects the source and reliability of each annotation. Deterministically resolved annotations are shown in **green**, while model-inferred annotations are shown in **orange** (R2, R3). LLM-based outputs are additionally prefixed with the symbol $\approx$ to indicate inferred meaning independent of color. Distinct type indicators differentiate annotation categories such as citations, abbreviations, and symbols. For mathematical symbols, the corresponding LaTeX representation is rendered as a small image and placed alongside the annotation text. All annotations are embedded as inline text elements rather than interactive overlays, ensuring compatibility with standard PDF viewers, screen readers, and print workflows. Upon completion, the system exports a structured JSON file containing annotation metadata, including type, page number, bounding box, content, and confidence level. This sidecar representation supports efficient interaction, filtering, and navigation in the accompanying viewer without requiring re-parsing of the document.

6 Interactive Viewer

GlossVis includes a lightweight, browser-based interactive viewer served locally via Flask. The viewer renders the annotated PDF using PDF.js and reads the sidecar JSON for annotation data, enabling real-time interaction without re-parsing the document. Three interaction mechanisms support exploratory reading.

Confidence filtering. Toggle buttons (*All*, *High*, *Medium*) filter the annotation panel in real time. Selecting *High* hides all medium-confidence annotations, directing attention to structurally guaranteed, author-defined definitions. The PDF rendering remains unchanged; only the annotation list is updated, preserving the reading context. This supports a progressive trust strategy, where readers can begin with all annotations and gradually focus on those they trust most.

Click-to-page navigation. Clicking an annotation in the panel scrolls the PDF view to the page where it appears. This supports a reading strategy in which readers scan the annotation panel for unfamiliar terms and navigate directly to their context in the document, effectively combining the Locate and Identify operations described in §3 within a single interaction.

Annotation search and sort. The panel supports live text search over annotation labels and allows sorting by page order (reading order) or by confidence level (most certain first). A live count displays the number of annotations matching the current filter. Sorting by confidence surfaces all green annotations first, enabling readers to quickly identify terms resolved with high structural certainty.

The interactive viewer is an optional component, and the annotated PDF remains a standalone artifact that can be read in any PDF viewer without it. All viewer state is computed client-side from the sidecar JSON, and no server-side processing occurs after the initial annotation run.

7 Visual Encoding

The source of each annotation (deterministic or model-inferred) is conveyed to the reader through three distinct encoding channels, following established practices in accessible uncertainty visualization [20]. The primary channel is `color`. Deterministically resolved annotations are rendered in **green**, while model-inferred annotations are shown in **orange**. This encoding leverages the pre-attentive traffic-light association (green,=,verified; orange,=,caution) and typically requires no legend (R2). Because color alone is insufficient for readers with color vision deficiency, a second channel encodes the same distinction using a `prefix marker`. Model-inferred annotations are prefixed with $\approx$, while deterministic annotations carry no prefix. The absence of a prefix thus serves as an explicit signal, ensuring readability in monochromatic displays and print. A third channel uses a `type icon` placed at the margin gutter. Symbols such as ● for citations, □ for abbreviations, and ▷ for mathematical symbols indicate annotation type independently of confidence and support type-based filtering in the interactive viewer without relying on color. The chosen green–orange palette remains distinguishable under deuteranopia and protanopia simulations [8]. Annotation text is set at 5 pt, the smallest size that remains legible at standard display resolution, to maximize the number of annotations per margin column without crowding. Each annotation is connected to its trigger

[8] http://www.vischeck.com

token by a 0.3 pt hairline rule, providing a spatial link that preserves reading context without occluding the body text (R1).

8 Evaluation

We structure the evaluation around four research questions (RQs). (RQ1) How does GlossVis compare to zero-shot LLM inference for abbreviation and symbol resolution? (RQ2) Does in-context retrieval (RAG) reduce the accuracy gap relative to zero-shot, and how does it compare to the full GlossVis pipeline? (RQ3) What is the contribution of the deterministic stage (DIE), and how does annotation coverage vary across papers? (RQ4) Does GlossVis reduce cognitive load and improve the reading experience in practice?

8.1 Evaluation Setup

Benchmark. We construct a gold-standard corpus of 12 seminal papers spanning eight domains, including TSGAN [32], Transformer [46], BERT [12], ViT [13], ResNet [17], U-Net [40], PPO [41], GCN [25], DDPM [19], Faster R-CNN [38], Swin Transformer [31], and EfficientNet [44]. The corpus comprises 172 pages, 235 abbreviations (40 inline-defined and 195 used-only), 74 mathematical symbols, and 418 citation references. Ground truth is generated through a four-step human-in-the-loop pipeline: (1) automated candidate extraction with local context, (2) manual coverage verification, (3) candidate resolution using a 14B-parameter LLM (`qwen2.5:14b`), selected for its best resolution performance (Fig. 6)., and (4) expert correction to produce final definitions.

We evaluate LLMs in two settings, zero-shot and in-context, alongside GlossVis. In the in-context setting, the same are provided with document chunks (context) using our RAG configuration (top-2 chunks, 500-character segments, `all-MiniLM-L6-v2` embeddings). We consider a diverse set of open LLMs spanning multiple scales to support flexible, cost-effective reading. These include `qwen2.5:1.5b` [51], `llama3.2:3b` [15], `qwen2.5:3b` [51], `gemma3:4b` [45], `mistral:7b` [23], `llama3.1:8b` [15], `qwen2.5:7b` [51], and `qwen2.5:14b` [51]. The metrics include *ECE* (Eq. 3), *Token-F1* (Eq. 4), and *Detection P/R/F1*. We also report *Exact Match Accuracy* (EMA), which measures the fraction of predictions that exactly match the ground-truth expansion. *Latency* is reported as the total wall-clock time per paper or per term.

8.2 GlossVis vs. LLMs in Zero-shot setting (RQ1)

Figure 3 compares zero-shot performance of eight LLMs against GlossVis. Figure 4 shows the per-paper EMA distribution to reveal consistency. Zero-shot abbreviation EMA ranges from 0.129 (`llama3.2:3b`) to 0.298 (`qwen2.5:14b`). Even the largest model fails on most paper-specific abbreviations, as definitions for used-only terms (195 of 235, 83%) are absent from pretraining data. Scaling from 1.5B to 14B improves EMA by only 0.116 points, confirming that model scale alone cannot replace document-grounded extraction. Symbol resolution is uniformly poor across all zero-shot models (EMA = 0.006–0.162), as symbols require the paper-specific notation convention rather than generic mathematical interpretations. GlossVis achieves abbreviation EMA = 0.379 and Token-F1 = 0.336, outperforming all zero-shot models. GlossVis produces annotations with near-perfect detection precision. In contrast, zero-shot models generate outputs for every queried term, regardless of whether a valid document-specific definition exists. The per-paper box plot (Figure 4) shows that this advantage is consistent. GlossVis outperforms zero-shot models on 9 of 12 papers for models up to 3B parameters, and on 6–8 of 12 papers for 7–8B models. The differences are statistically significant (McNemar $p<0.001$). The DIE component plays a central role in resolving inline-defined abbreviations. It achieves an EMA of 0.615 with a detection precision of 0.833, which is 2.3× higher than the best zero-shot model (0.298). Table 1 reports all metrics (EMA, TF1, ECE) across conditions. Token-F1 closely tracks EMA (r=0.98) and is included for completeness. EMA remains the primary metric as it reflects the exact annotation presented to the reader.

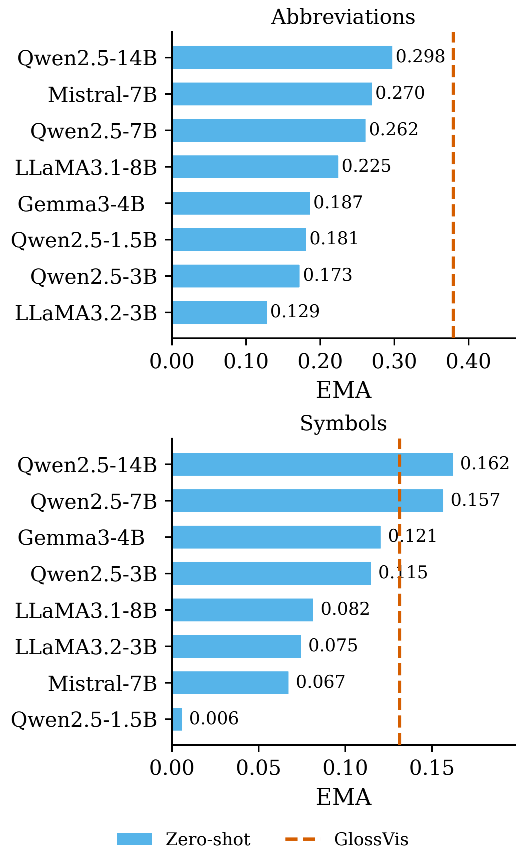

Fig. 3: LLM in zero-shot setting EMA for abbreviations (top) and symbols (bottom), sorted in ascending order.

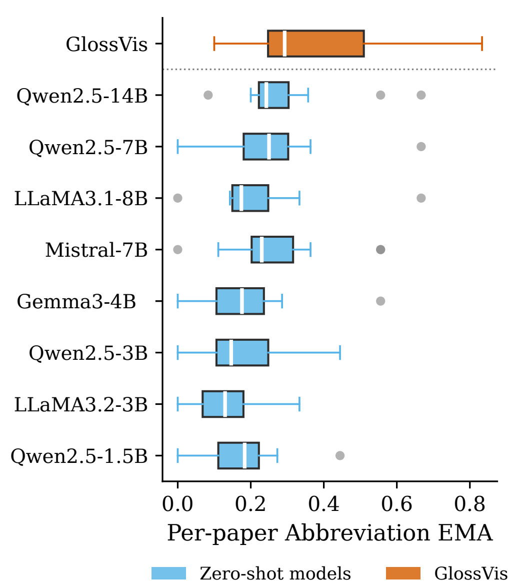

Fig. 4: Per-paper abbreviation EMA distributions for each LLM in zero-shot setting and GlossVis.

8.3 In-Context LLMs Reduce the Performance Gap? (RQ2)

We evaluate whether document context alone reduces the performance gap and how model size interacts with retrieval in the full GlossVis system. Figure 5 shows the LLM in zero-shot (ZS) set-

Table 1: Comparison across baselines by model size (macro-average). ⋆ production default, ‡ ablation (fallback swapped), † inline-defined terms only. Orange indicates the lowest performance.

Condition	Model	Size	Resolution Quality EMA ↑	TF1 ↑	ECE ↓	Lat/term ↓
Small-scale						
Zero-shot	Qwen2.5-1.5B	1.5B	0.181	0.193	0.794	0.35s
RAG-only	Qwen2.5-1.5B	1.5B	0.256	0.338	0.728	0.60s
GlossVis‡	**Qwen2.5-1.5B**	**1.5B**	**0.312**	**0.293**	**0.265**	**0.77s†**
Default (production)						
Zero-shot	Gemma3-4B	4B	0.187	0.206	0.795	0.55s
RAG-only	Gemma3-4B	4B	0.305	0.424	0.695	0.89s
GlossVis⋆	**Gemma3-4B**	**4B**	**0.379**	**0.336**	**0.258**	**1.10s†**
Large-scale (ceiling)						
Zero-shot	Qwen2.5-14B	14B	0.298	0.321	0.676	0.67s
RAG-only	Qwen2.5-14B	14B	0.462	0.568	0.422	1.69s
GlossVis‡	**Qwen2.5-14B**	**14B**	**0.433**	**0.369**	**0.211**	**1.95s†**

ting to RAG in-context setting progression. In-context retrieval improves abbreviations and symbols EMA for all models, from the zero-shot range of 0.129–0.298 to 0.210–0.462 under RAG. The best LLM with RAG (`qwen2.5:14b`, EMA,=,0.462) exceeds the EMA of GlossVis. However, this gain comes with a substantial increase in model size and latency. Moreover, LLM+RAG systems assign equal confidence to all annotations and cannot distinguish reliable from uncertain resolutions. GlossVis explicitly partitions annotations into HIGH (DIE, $P(a)$=0.95, acc = 0.615) and MEDIUM (RAG, $P(a)$=0.65, acc = 0.526) tiers, making reliability legible through color encoding. Citation resolution is not evaluated under zero-shot setting, as LLMs lack access to the document's reference list. In-context RAG achieves citation EMA in the range of 0.482–0.640 across models, with `gemma3:4b` (0.636) and `qwen2.5:14b` (0.640) performing best. GlossVis achieves the highest citation EMA of 0.649. The citation detection performance of the full s (P = 0.833, R = 0.725) reflects cases where citation marker formats vary across papers and are not always captured by the reference-index lookup.

We conduct an ablation study to quantify the tradeoff between latency, accuracy, and model size in GlossVis across different LLM backends, as shown in Figure 6. Smaller models such as `qwen2.5:1.5b` (0.77 s/term) reduce latency with a moderate drop in EMA, while larger models such as `qwen2.5:14b` (1.95 s/term) improve EMA by 5 points but roughly double processing time. Gemma3-4B (1.1 s/term) provides the best balance between accuracy, latency, and parameter size among the evaluated models.

8.4 Contribution of the Deterministic Component (RQ3)

We evaluate the deterministic resolution component of GlossVis by completely disabling the LLM fallback. Figure 7 shows the distribution

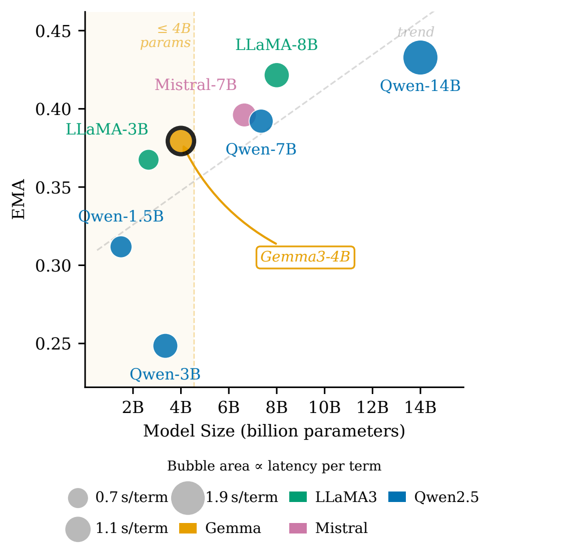

Fig. 6: Tradeoff between EMA, latency (s/term), and model size for different LLM backends in GlossVis.

per-paper of abbreviations resolved by DIE, resolved by LLM-RAG, and unsolved in the full GlossVis pipeline. Figure 8 shows the detection Precision/Recall across glossary element types for DIE vs. full GlossVis system. For inline-defined abbreviations, DIE achieves an EMA of 0.615 with a detection precision of 0.833. The low recall (R = 0.243) reflects the limited scope of the DIE stage. It applies only when a definitional pattern appears in the document text. The proportion of abbreviations resolved by DIE varies substantially across papers. GlossVis extends this to P = 1.000 and R = 0.672 on abbreviations, and achieves almost perfect detection on symbols. The abbreviation recall gap (0.328 unannotated) reflects cases where the LLM returns empty output, fails ILV or candidates are filtered to suppress incidental tokens, trading recall for higher precision. In these cases, withholding annotations is preferable to displaying hallucinated expansions. Further, we evaluate the alignment between assigned confidence levels and empirical accuracy. The HIGH tier ($P(a)$=0.95) achieves an accuracy of 0.667 on inline-defined terms, while the MEDIUM tier ($P(a)$=0.65) achieves an accuracy of 0.365 on RAG-resolved terms. The ECE results in Table 1 show that GlossVis achieves ECE = 0.258, compared to 0.763 (zero-shot) and 0.621 (RAG-only), a threefold improvement validating the two-tier confidence design.

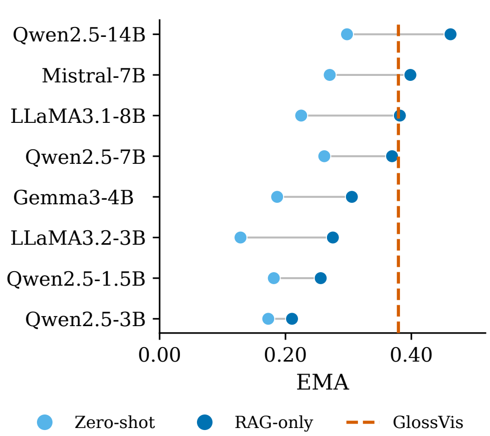

Fig. 5: LLM mean EMA for abbreviation and symbol resolution. Sky-blue circles denote zero-shot, and blue circles denote in-context (RAG-only). Lines connect each model's zero-shot and in-context performance.

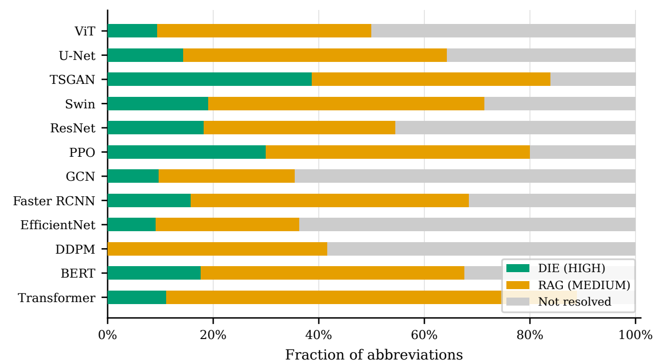

Fig. 7: Per-paper abbreviation coverage in the full GlossVis system. Green denotes DIE-resolved annotations (HIGH confidence), orange denotes RAG-resolved annotations (MEDIUM confidence), and gray denotes unannotated cases where the LLM returns empty or low-confidence outputs.

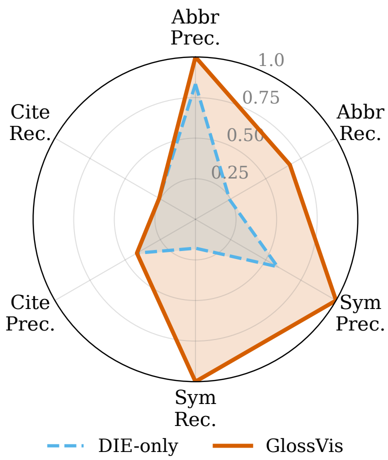

Fig. 8: Precision (Prec.) and recall (Rec.) for DIE-only (sky blue, dashed) and GLOSSVIS (red) across abbreviations, citations, and symbols.

Table 2: Human evaluation results. Ratings are on a 1–5 Likert scale. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$ vs. neutral midpoint (Wilcoxon signed-rank).

Metric	Mean	SD	Median	95% CI	Sig.
Quality metrics					
Semantic accuracy	3.46	0.97	3.0	±0.30	**
Citation relevance	3.59	0.82	4.0	±0.26	***
Symbol recognition	3.77	0.93	4.0	±0.29	***
Non-overlapping	3.69	0.98	4.0	±0.31	***
Color-coding clarity	4.00	0.95	4.0	±0.30	***
Cognitive Load metrics					
Anchor alignment	4.10	0.88	4.0	±0.28	***
Context-switch reduce	3.90	0.82	4.0	±0.26	***
Mental effort reduce	3.56	0.94	4.0	±0.29	**
Perceived efficiency	3.69	0.80	4.0	±0.25	***
Overall metrics					
Overall satisfaction	3.85	0.81	4.0	±0.26	***

8.5 Human Evaluation (RQ4)

We conducted a within-subjects user study with nine domain-specific evaluators. Each evaluator read one or more GLOSSVIS-annotated PDFs and completed a short questionnaire for each paper. The questionnaire included nine Likert-scale questions, each with five response options. These were grouped into *Annotation Quality* (semantic accuracy, citation relevance, symbol recognition, visual clarity, and color-coding clarity) and *Cognitive Load* (anchor alignment, reduced context switching, mental effort, and perceived efficiency). It also included a three-option preference question. We received $N=39$ valid responses covering 39 unique papers across diverse domains. Internal consistency is acceptable, with $\alpha_{\text{quality}}=0.791$ and $\alpha_{\text{cog}}=0.709$ (both ≥ 0.70 [5]). Figure 9 and Table 2 report per-item statistics. All five quality metrics are significantly above the neutral midpoint of 3 (Wilcoxon signed-rank test, all $p<0.01$). Color-coding clarity scores highest (mean = 4.00, $p<0.001$), validating the triple encoding (§7). Semantic accuracy scores lowest among the quality items (mean = 3.46, $p<0.01$).

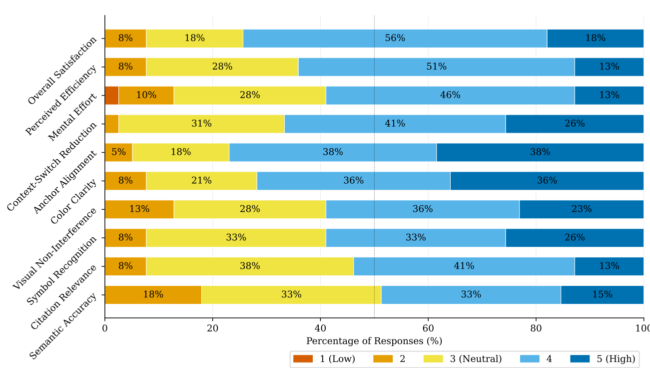

Fig. 9: Mean ratings with 95% confidence intervals for all nine Likert items, grouped by construct ($N=39$ responses from 9 evaluators).

All four cognitive load items are significantly above the neutral midpoint ($p<0.01$). Anchor alignment scores highest (mean = 4.10, $p<0.001$), indicating that spatial co-location of annotations with their trigger tokens satisfies requirement R1. Context-switching reduction (3.90) and perceived efficiency (3.69) both suggest that readers experience fewer interruptions. The composite cognitive load score is 3.81 ± 0.63 ($p<0.001$ vs. neutral). Spearman correlation analysis reveals a significant negative relationship between domain familiarity and symbol recognition ($\rho=-0.478$, $p<0.01$), indicating that expert readers are more likely to identify symbol annotation errors than novices. Citation relevance and perceived efficiency show weaker familiarity effects ($\rho=-0.333$ and -0.383, both $p<0.05$). In contrast, cognitive load metrics show no significant effect of familiarity ($p>0.50$), suggesting that the layout and readability benefits are consistent across different levels of domain expertise. A total of 62% of respondents prefer reading with GLOSSVIS over a standard PDF, 33% are uncertain, and 5% prefer the standard format. Overall satisfaction is 3.85 ± 0.81 ($p<0.001$ vs. neutral). Evaluators noted that the **green**/**orange** distinction supports on-the-fly trust calibration without requiring external references.

9 Discussion and Future Work

The results validate the core design principle of GLOSSVIS, which prioritizes document-grounded extraction over model inference. Document-grounded extraction achieves an EMA of 0.615 on inline-defined abbreviations, more than double the best zero-shot baseline (0.298), with negligible latency. Retrieval context contributes more to accuracy than model scale, as a 1.5B model with RAG (EMA = 0.256) approaches the performance of a 14B model with zero-shot (EMA = 0.298) while using 9× fewer parameters. For resource-constrained deployment, improving retrieval is more effective than scaling the inference model. The two-tier confidence design is also well supported, with GLOSSVIS achieving an ECE of 0.258 compared to 0.763 for zero-shot methods, a threefold improvement in calibration. Evaluators further reported that the color encoding helped them assess reliability without leaving the document. In the absence of an in-document definition, annotations are omitted to avoid introducing incorrect explanations. Several limitations remain. Recall is constrained by the availability of explicit definitions in the source document. While the structural validator reduces errors, it does not eliminate semantically plausible mismatches. The system also does not support scanned documents. Future work includes conducting controlled studies to evaluate reading efficiency and trust. It also involves incorporating a lightweight verification stage for model outputs, extending symbol coverage to multi-character expressions, and improving confidence calibration across a wider range of domains.

10 Conclusion

We present GLOSSVIS, a system for annotating glossary information directly within document margins while preserving spatial context. The system distinguishes between author-defined and model-inferred content through uncertainty-aware visual encoding. Our evaluation shows that GLOSSVIS outperforms or matches zero-shot and in-context LLMs on glossary element resolution. A user study further demonstrates reduced cognitive load and a clear preference for in-document annotations. These findings suggest that combining document-grounded extraction with uncertainty-aware visualization enables AI-assisted reading that is both reliable and effective.

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